



Overview

This year's technical performance chapter features analysis of the technical progress in AI during 2022. Building on previous reports, this chapter chronicles advancement in computer vision, language, speech, reinforcement learning, and hardware. Moreover, this year this chapter features an analysis on the environmental impact of AI, a discussion of the ways in which AI has furthered scientific progress, and a timeline-style overview of some of the most significant recent AI developments.

Chapter Highlights

Performance saturation on traditional benchmarks.

AI continued to post state-of-the-art results, but year-over-year improvement on many benchmarks continues to be marginal. Moreover, the speed at which benchmark saturation is being reached is increasing. However, new, more comprehensive benchmarking suites such as BIG-bench and HELM are being released.

AI systems become more flexible.

Traditionally AI systems have performed well on narrow tasks but have struggled across broader tasks. Recently released models challenge that trend; BEiT-3, PaLI, and Gato, among others, are single AI systems increasingly capable of navigating multiple tasks (for example, vision, language).

AI is both helping and harming the environment.

New research suggests that AI systems can have serious environmental impacts. According to Luccioni et al., 2022, BLOOM's training run emitted 25 times more carbon than a single air traveler on a one-way trip from New York to San Francisco. Still, new reinforcement learning models like BCOOLER show that AI systems can be used to optimize energy usage.

Generative AI breaks into the public consciousness.

2022 saw the release of text-to-image models like DALL-E 2 and Stable Diffusion, text-to-video systems like Make-A-Video, and chatbots like ChatGPT. Still, these systems can be prone to hallucination, confidently outputting incoherent or untrue responses, making it hard to rely on them for critical applications.

Capable language models still struggle with reasoning.

Language models continued to improve their generative capabilities, but new research suggests that they still struggle with complex planning tasks.

The world's best new scientist ... AI?

AI models are starting to rapidly accelerate scientific progress and in 2022 were used to aid hydrogen fusion, improve the efficiency of matrix manipulation, and generate new antibodies.

AI starts to build better AI.

Nvidia used an AI reinforcement learning agent to improve the design of the chips that power AI systems. Similarly, Google recently used one of its language models, PaLM, to suggest ways to improve the very same model. Self-improving AI learning will accelerate AI progress.



The technical performance chapter begins with an overview of some of the most significant technical developments in AI during 2022, as selected by the AI Index Steering Committee.

2.1 What's New in 2022: A Timeline

Feb. 2, 2022

DeepMind Releases AlphaCode

AlphaCode, an AI system that writes computer programs at a competitive level, achieves a rank within the top 54% of participants in a human programming competition. This represents an improvement on the more complex problem-solving tasks with which AI has traditionally struggled.

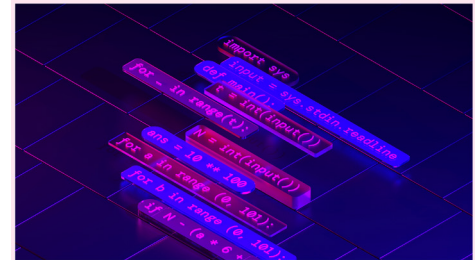


Figure 2.1.1

Feb. 16, 2022

DeepMind Trains Reinforcement Learning Agent to Control Nuclear Fusion Plasma in a Tokamak

Nuclear fusion is a potential source of clean, limitless energy, but producing such energy in tokamaks is difficult due to a lack of experimental data. DeepMind simulated optimal tokamak management, an example of how AI can accelerate science and combat climate change.

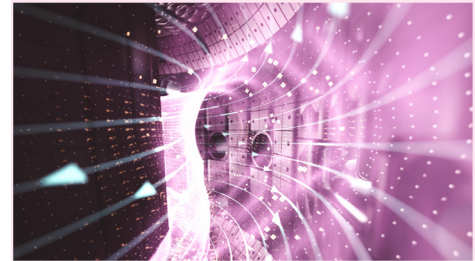


Figure 2.1.2

March 10, 2022

IndicNLG Benchmarks Natural Language Generation for Indic Languages

An international research collective launches IndicNLG, a collection of datasets for benchmarking natural language generation for 11 Indic languages. The creation of IndicNLG increases the potential for AI systems to generate language in more diverse, non-English linguistic settings.

Task	Languages	Communicative Intent	Input Type	Size
Biography Generation	L-{gu, mr}	One-sentence biography	key-value pairs	57K
Headline Generation	L	News article headlines	news article	1.31M
Sentence Summarization	L	Synonymous compact sentence	sentence	431K
Paraphrase Generation	L	Synonymous sentence	sentence	5.57M
Question Generation	L	Question leading to answer given context	context-answer pairs	1.08M

Figure 2.1.3



March 24, 2022

Meta AI Releases Make-A-Scene

Make-A-Scene is a text-to-image AI model that enables users to generate images through text. Make-A-Scene is one of many text-to-image models released in 2022.



Figure 2.1.4

April 5, 2022

Google Releases PaLM

Google's AI team trains one of the world's largest language models, PaLM. Made up of 540 billion parameters, PaLM reinforces the belief that researchers can improve performance on large language models by simply training them on more data.

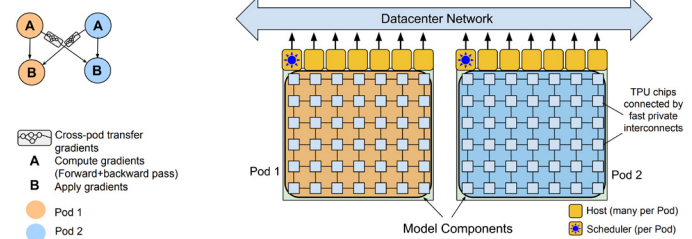


Figure 2.1.5

April 13, 2022

OpenAI Releases DALL-E 2

DALL-E 2, a text-to-image AI system that can create realistic art and images from textual descriptions, is released to the public, igniting a generative AI craze.

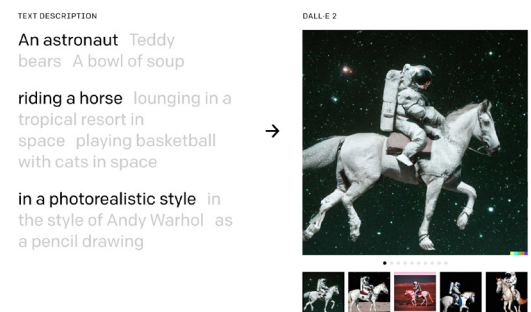


Figure 2.1.6

May 12, 2022

DeepMind Launches Gato

Gato is a new reinforcement learning agent capable of doing a wide range of tasks such as robotic manipulation, game playing, image captioning, and natural language generation. The release of such models suggests that AI systems are becoming better at generalization.



Figure 2.1.7

Computer vision is the subfield of AI that teaches machines to understand images and videos. Computer vision technologies have a variety of important real-world applications, such as autonomous driving, crowd surveillance, sports analytics, and video-game creation. This section tracks progress in computer vision across several different task domains which include: (1) image classification, (2) face detection and recognition, (3) deepfake detection, (4) human pose estimation, (5) semantic segmentation, (6) medical image segmentation, (7) object detection, (8) image generation, and (9) visual reasoning.

2.2 Computer Vision—Image

Image Classification

Image classification is the ability of machines to categorize objects in images (Figure 2.2.1).

ImageNet

ImageNet is one of the most widely used benchmarks for image classification. This dataset includes over 14 million images across 20,000 different object categories such as “strawberry” or “balloon.” Performance on ImageNet is measured through various accuracy metrics. Top-1 accuracy measures the degree to which the top prediction generated by an image classification model for a given image actually matches the image’s label.

As of 2022, the best image classification system on ImageNet has a top-1 accuracy rate of 91.0% (Figure 2.2.2). Although the current image classification capabilities of state-of-the-art systems is 27.7 percentage points better than a decade ago, last year saw a very marginal 0.1 percentage point improvement in classification accuracy.

A Demonstration of Image Classification

Source: Krizhevsky et al., 2012

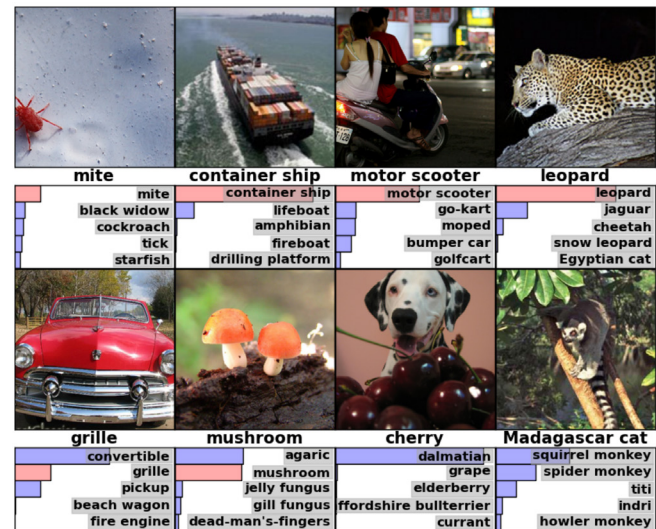


Figure 2.2.1

ImageNet Challenge: Top-1 Accuracy

Source: Papers With Code, 2022; arXiv, 2022 | Chart: 2023 AI Index Report

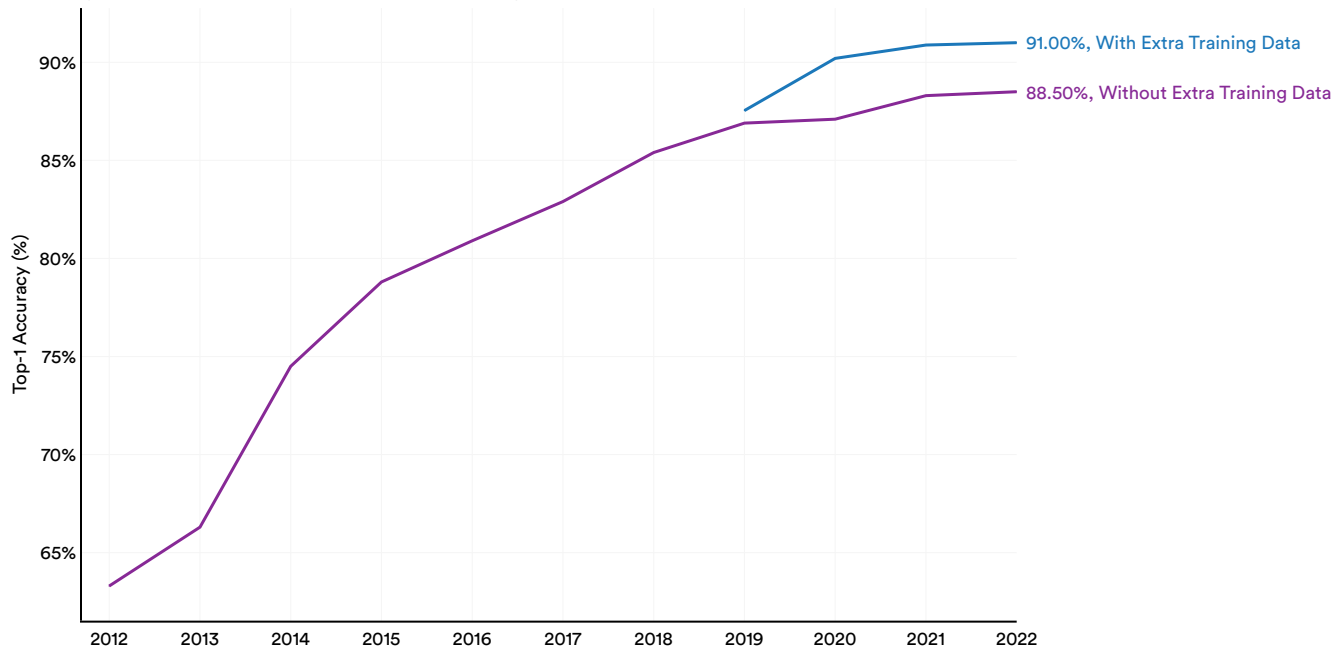


Figure 2.2.2

Face Detection and Recognition

Facial detection and recognition is the ability of AI systems to identify faces or individuals in images or videos (Figure 2.2.3). Currently, many facial recognition systems are able to successfully identify close to 100% of faces, even on challenging datasets (Figure 2.2.4).

A Demonstration of Face Detection and Recognition

Source: [Forbes, 2020](#)

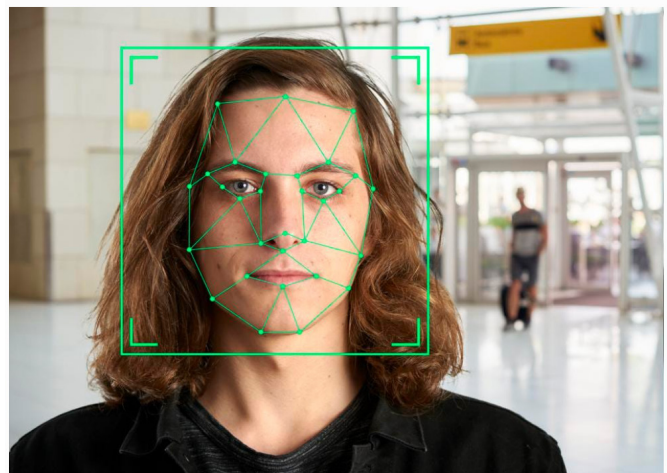


Figure 2.2.3



National Institute of Standards and Technology (NIST) Face Recognition Vendor Test (FRVT): Verification Accuracy by Dataset

Source: National Institute of Standards and Technology, 2022 | Chart: 2023 AI Index Report

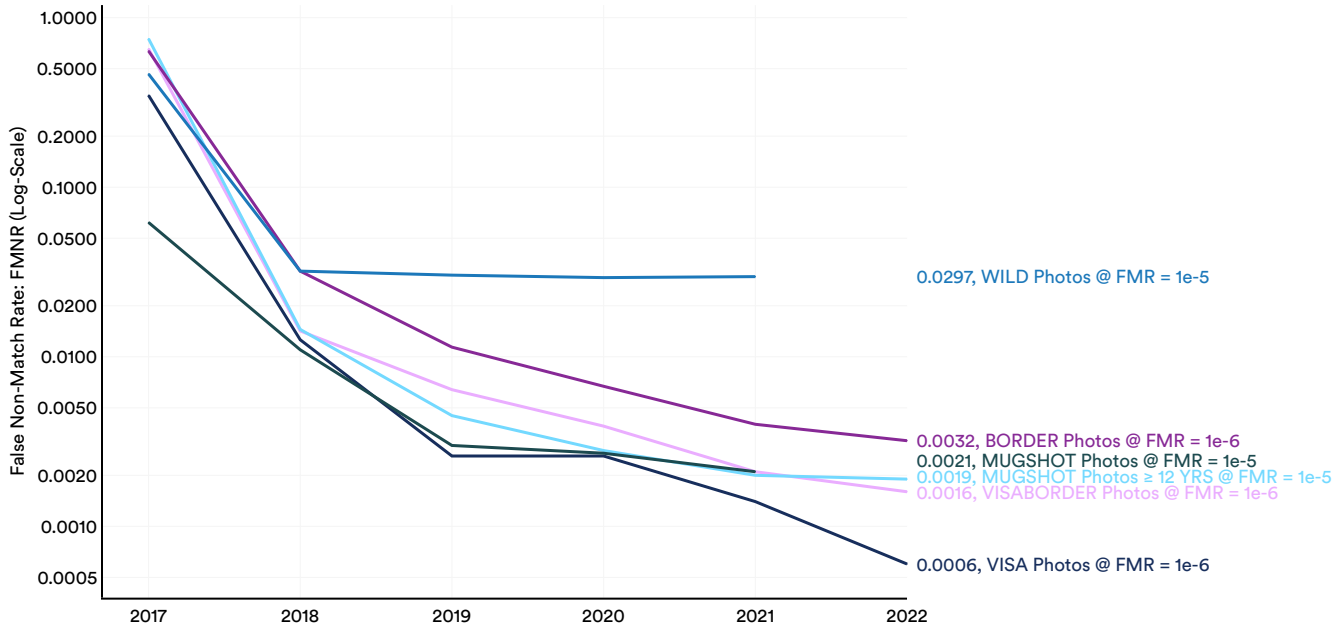


Figure 2.2.4

National Institute of Standards and Technology Face Recognition Vendor Test (FRVT)

Progress on facial recognition can be tracked through the [National Institute of Standards and Technology's Face Recognition Vendor Test](#). This test tracks how well different facial recognition algorithms perform on various homeland security tasks, such as identification of child trafficking victims and cross-verification of visa images, among

others. Facial detection capacity is measured by the false non-match rate (FNMR), otherwise known as error rate, which is the rate at which a model fails to match the face in an image to that of a person.

As of 2022, the top-performing models on all of the FRVT datasets, with the exception of WILD Photos, each posted an error rate below 1%, and as low as a 0.06% error rate on the VISA Photos dataset.

Object Detection

The challenge of identifying and localizing objects within an image or video is known as object detection (Figure 2.2.13).

Common Objects in Context (COCO)

Microsoft's Common Objects in Context (COCO) object detection dataset has over 80 object categories in 328,000 images. Several accuracy metrics are used to measure progress on COCO. This section considers mean average precision (mAP50).

Since 2015, state-of-the-art detectors have improved by 26 percentage points. The top model in 2022, EVA, was the result of a Chinese academic research collaboration.

A Demonstration of Object Detection

Source: [Rizzoli, 2023](#)

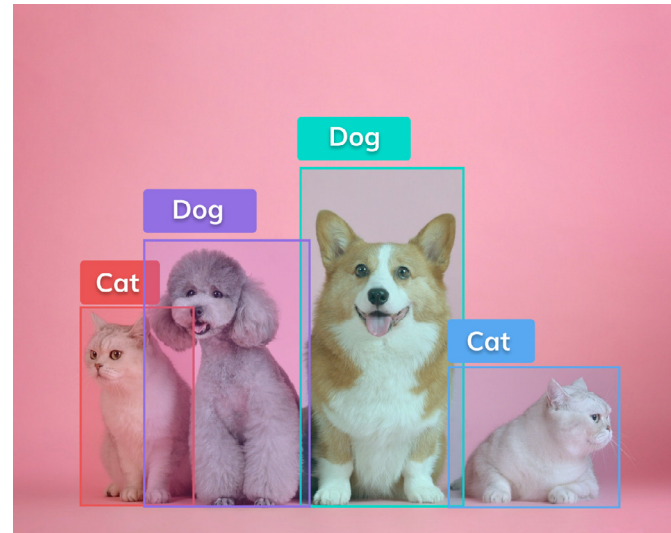


Figure 2.2.13

COCO: Mean Average Precision (mAP50)

Source: Papers With Code, 2022; arXiv, 2022 | Chart: 2023 AI Index Report

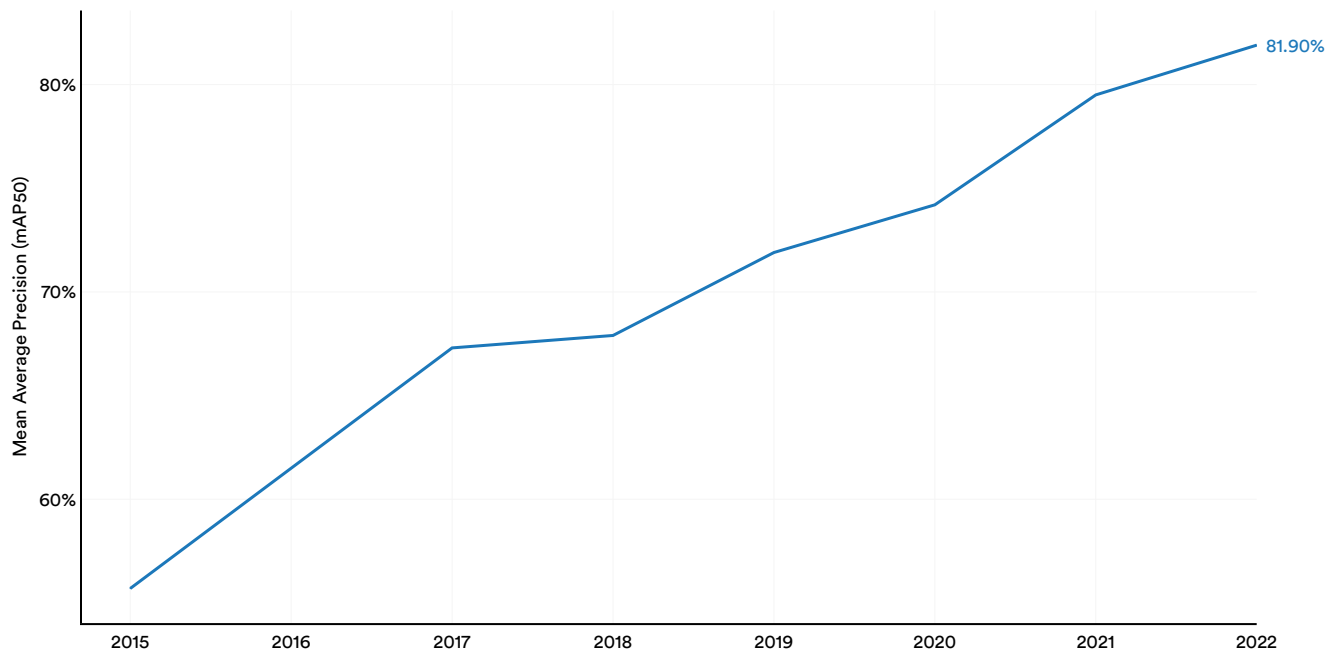


Figure 2.2.14

Narrative Highlight:

A Closer Look at Progress in Image Generation

Figure 2.2.17 tracks the progress of facial image generation over time, with the final image being generated by Diffusion-GAN, the model that posted the 2022 state-of-the-art score on STL-10.

GAN Progress on Face Generation

Source: [Goodfellow et al., 2014](#); [Radford et al., 2016](#); [Liu and Tuzel, 2016](#); [Karras et al., 2018](#); [Karras et al., 2019](#); [Goodfellow, 2019](#); [Karras et al., 2020](#); [Vahdat et al., 2021](#); [Wang et al., 2022](#).

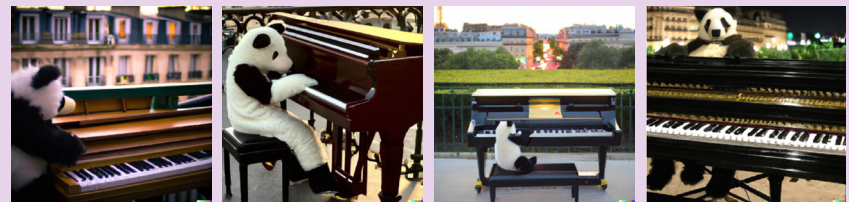


Figure 2.2.17

In the last year, text-to-image generation broke into the public consciousness with the release of models such as OpenAI’s [DALL-E 2](#), Stability AI’s [Stable Diffusion](#), Midjourney’s [Midjourney](#), Meta’s [Make-A-Scene](#), and Google’s [Imagen](#). With these systems, users can generate images based on a text prompt. Figure 2.2.18 juxtaposes the images generated by DALL-E 2, Stable Diffusion, and Midjourney, three publicly accessible AI text-to-image systems, for the same prompt: “a panda playing a piano on a warm evening in Paris.”

Images Generated by DALL-E 2, Stable Diffusion and Midjourney

Source: AI Index, 2022



a. DALL-E 2



b. Stable Diffusion



c. Midjourney

Figure 2.2.18

Narrative Highlight:

A Closer Look at Progress in Image Generation (cont'd)

Of all the recently released text-to-image generators, Google's Imagen performs best on the COCO benchmark (Figure 2.2.19)². This year, the [Google researchers](#) who created Imagen also released a more difficult text-to-image benchmark, DrawBench, designed to challenge increasingly capable text-to-image models.

Notable Text-to-Image Models on MS-COCO 256 × 256 FID-30K: Fréchet Inception Distance (FID) Score

Source: Saharia et al., 2022 | Chart: 2023 AI Index Report

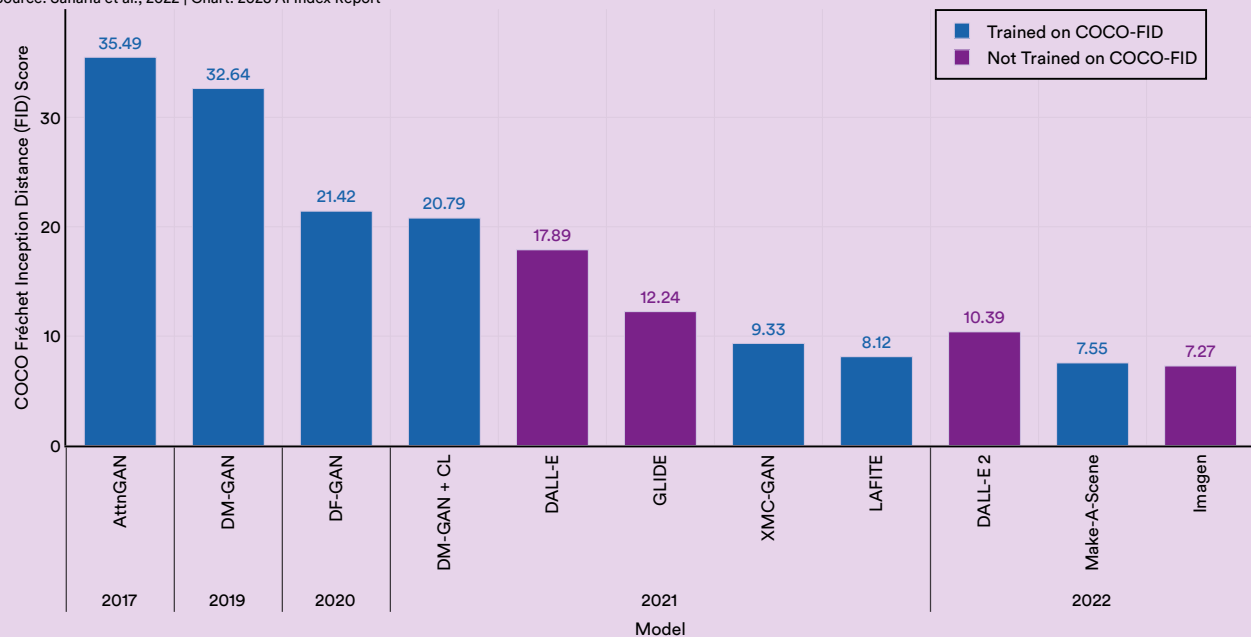


Figure 2.2.19

² The COCO benchmark, first launched in 2014, includes 328,000 images with 2.5 million labeled instances. Although it is typically used for object detection tasks, researchers have also deployed it for image generation.